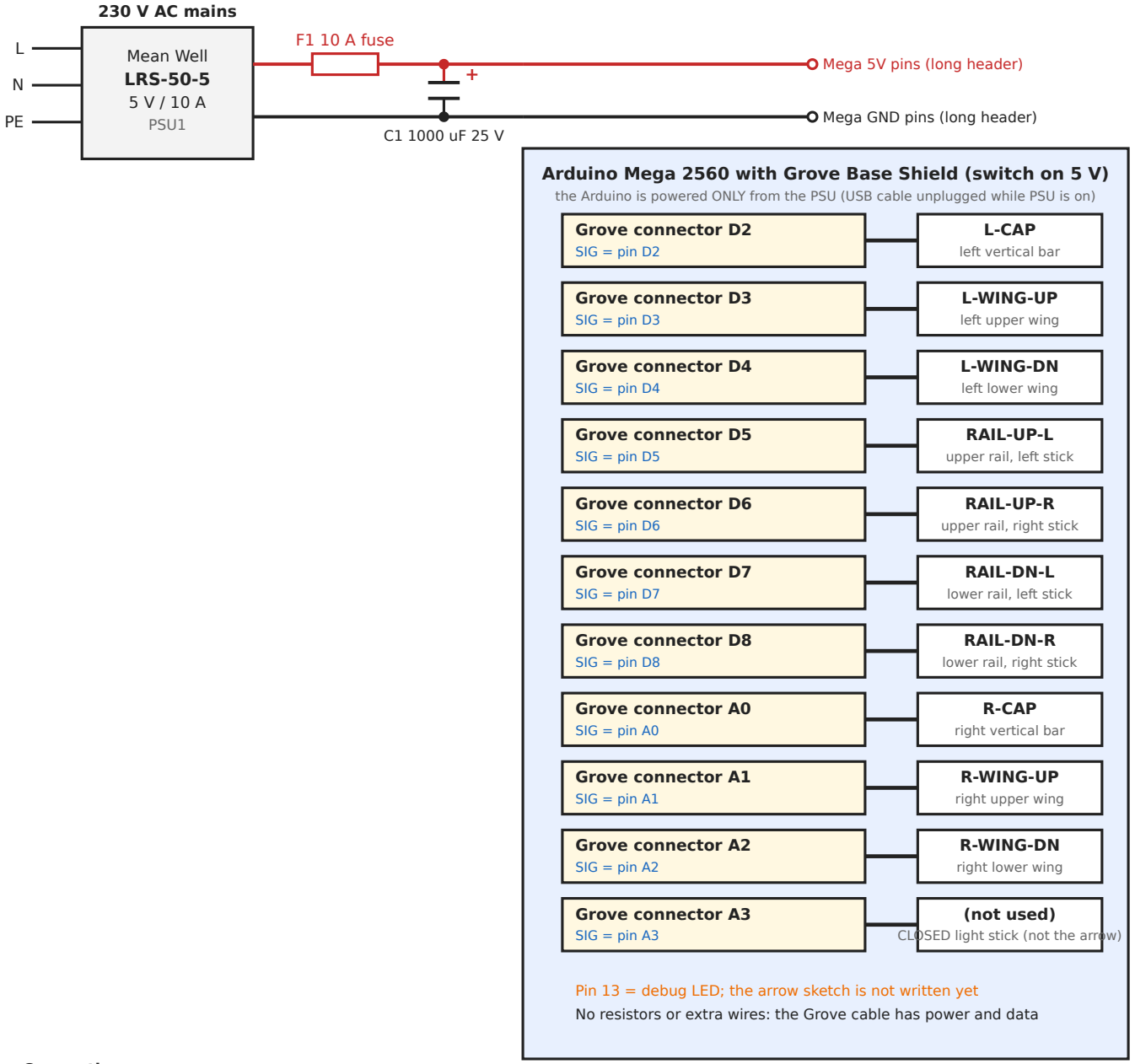
Stick arrow - wiring (10 named Grove sticks plugged unmodified into the Grove Base Shield)



Connections

1. Each stick is plugged with its own unmodified Grove cable into the connector shown: L-CAP D2 ... RAIL-DN-R D8,
R-CAP A0, R-WING-UP A1, R-WING-DN A2; the connector A3 carries the sign's CLOSED light stick, not part of the arrow (the stick names are in layout-diagram
2. The shield 3V3/5V switch must be on 5 V. The stick power (red/black wires) goes through the shield and the Grove connectors.
3. PSU V+ -> 10 A fuse F1 -> the Mega 5V pins; PSU V- -> the Mega GND pins (use the 5V and GND pins at the end of the long
double header, away from the shield; use both 5V pins and both GND pins, thick wire 1.5 mm2). C1 across the supply wires.
4. Power the Arduino ONLY from the PSU: do not plug the USB cable while the PSU is on (upload the sketch first with USB, unplug, then power).
5. No Serial Monitor output without USB; the pin 13 LED shows that the program runs.
6. Feel the shield, the connectors and the wires after running all LEDs white for a minute: if anything gets warm, switch off and lower the brightness.
7. The mains side of the supply must be in an enclosure and wired by a person who knows how (nothing here is mains).